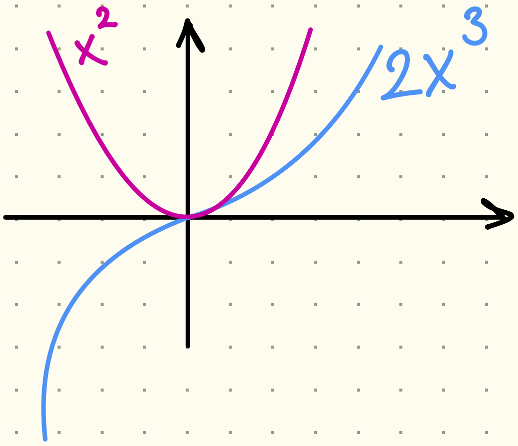
# LIMITI (ES)

es) $\lim_{x \to -\infty} \underline{2x^3} + \underline{x^2} - 3?$




| $x$ | $x^2$ | $x^3$ |
|---|---|---|
| 10 | $10^2$ | $10^3$ |
| 100 | $10^4$ | $10^6$ |
| 1000 | $10^6$ | $10^9$ |

• $\lim_{x \to -\infty} 2x^3 + x^2 - 3$

$$= \lim \underbrace{x^3}_{\to -\infty} \underbrace{\Big( \underbrace{2}_{\to 2} + \underbrace{\frac{1}{x}}_{\to 0} - \underbrace{\frac{3}{x^3}}_{\to 0} \Big)}_{\to 2} = -\infty$$

es) $\lim_{x \to +\infty} \frac{\arctan x}{x - 3x^2}$

$$x - 3x^2 = x(1 - 3x) \to -\infty$$

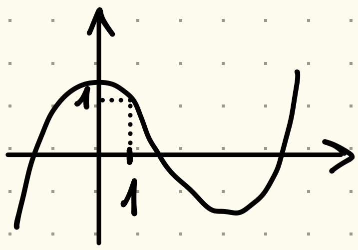
POSSO ANCHE RACCOGLIERE PER $x^2$

$$= \underbrace{x^2}_{-\infty} \underbrace{\left( \frac{1}{x} - 3 \right)}_{3} \to -\infty$$

es) $\lim_{x\to 1}\frac{\sin x}{x-1}$ → SI ANALIZZANO NUM. E DEN. SINGOLARM.

N: $\sin 1$

D: $1-1=0$




• $\lim_{x\to 1^-}\frac{\overbrace{\sin x}^{\to\ \sin 1>0}}{\underbrace{x-1}_{\to\ 0^-}} = -\infty$

• $\lim_{x\to 1^+}\frac{\overbrace{\sin x}^{\to\ \sin 1}}{\underbrace{x-1}_{\to\ 0^+}}$

QUINDI $\nexists \lim_{x\to 1}\frac{\sin x}{x-1}$

es) $\lim_{x\to+\infty}\frac{x^2+3x}{x-3x^2}$

• $\lim_{x\to+\infty}\frac{\overbrace{x^2+3x}^{\to\ +\infty}}{\underbrace{x-3x^2}_{\to\ -\infty}}$

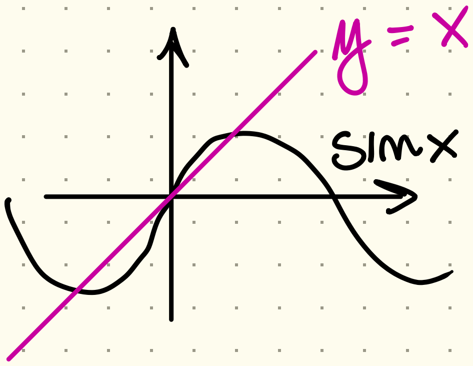
• $\frac{x^2+3x}{x-3x^2} = \frac{x^2(1+3/x)}{x^2(1/x-3)} = \frac{1+3/x \to 1}{1/x-3 \to -3}$

$\downarrow x\to-\infty$

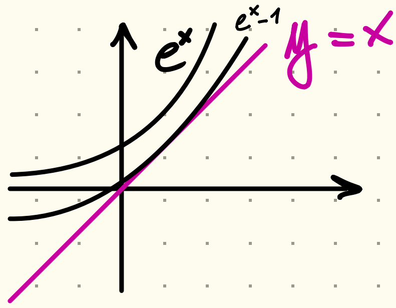
$-\frac{1}{3}$

# • FORMA INDETERMINATA / LIMITI NOTEVOLI

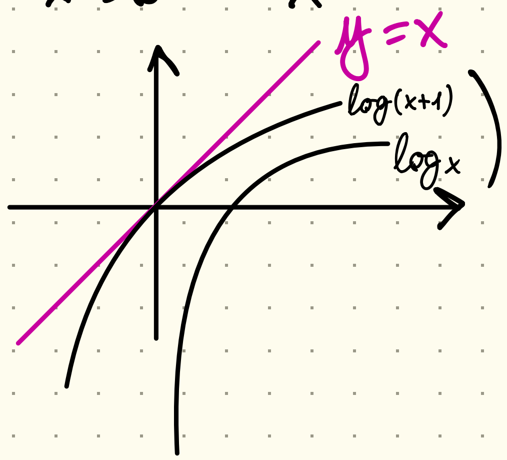
- $\lim_{x \to 0} \frac{\sin x}{x} = 1$




- $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$




- $\lim_{x \to 0} \frac{\log(x+1)}{x} = 1$




$\log_e$ → IN BASE $e$

- $$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$$

- $$\lim_{x\to 0} \frac{x-\cos x}{x^3} = \frac{1}{6}$$

- $$\lim_{x\to +\infty} \frac{a^x}{x^n} = +\infty \qquad \begin{matrix} a>1 \\ n\in\mathbb{N} \end{matrix}$$

- $$\lim_{x\to +\infty} \frac{x^n}{\log x} = +\infty$$

- $$\lim_{x\to -\infty \to \pm\infty \to 0} x^n e^x = 0$$

- $$\lim_{x\to 0} x \log x = 0$$

- $$\lim_{x\to +\infty} \log x - x = -\infty$$

## • Limite Funzione Composta

- $$\lim_{x\to x_0} g(f(x)) = \lim_{y\to y_0} g(y)$$

- $$\lim_{x\to x_0} f(x) = y_0$$

Per la composta, svolgo l'interna e poi l'esterna

$$\cdot \lim_{x \to 1} \frac{\sin(x-1)}{x-1} = \lim_{y \to y_0} \frac{\sin y}{y} = 1$$

$$g(y) = \frac{\sin y}{y} \qquad y = \underbrace{x-1}_{\emptyset}$$

$$f(x) = x-1$$

$$\lim_{x \to 1} x - 1 = \emptyset = y_0$$